Infinite Series Worksheet (9.3) — Solutions

Type A: Geometric Series

A geometric series has the form

$$a + ar + ar^2 + \cdots + ar^{k-1} + \cdots,$$

where each term is obtained by multiplying the previous term by the same number r.

How to find a and r:

1. $a =$ **the first term** of the series. Plug in the *starting* value of k (which may be $k = 0$, $k = 1$, $k = 2$, …).
2. $r =$ **(second term) ÷ (first term)**. To be sure the series is geometric, also check that the ratio of *any* two consecutive terms, $\dfrac{u_{k+1}}{u_k}$, is the same constant r.
3. If $|r| < 1$, the series converges and its sum is $\dfrac{a}{1-r}$. If $|r| \geq 1$, the series diverges.

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1. $\displaystyle\sum_{k=0}^{\infty} \frac{7}{3^k} = 7 + \frac{7}{3} + \frac{7}{3^2} + \cdots + \frac{7}{3^k} + \cdots$

Find a. The series starts at $k = 0$, so the first term is

$$a = \frac{7}{3^0} = \frac{7}{1} = 7.$$

Find r. The second term ($k = 1$) is $\dfrac{7}{3}$, so

$$r = \frac{\text{second term}}{\text{first term}} = \frac{7/3}{7} = \frac{1}{3}.$$

Check with the general term: $\dfrac{u_{k+1}}{u_k} = \dfrac{7/3^{k+1}}{7/3^k} = \dfrac{3^k}{3^{k+1}} = \dfrac{1}{3}$ for every k. ✓

Converge or diverge? $|r| = \frac{1}{3} < 1$, so the series converges and

$$\sum_{k=0}^{\infty} \frac{7}{3^k} = \frac{a}{1-r} = \frac{7}{1-\frac{1}{3}} = \frac{7}{\frac{2}{3}} = 7 \cdot \frac{3}{2} = \frac{21}{2}.$$

Answer: $\boxed{\text{converges, sum} = \dfrac{21}{2}}$

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2. $\displaystyle\sum_{k=0}^{\infty} \frac{(-1)^k\, 6}{5^k} = 6 - \frac{6}{5} + \frac{6}{5^2} - \cdots + \frac{(-1)^k\, 6}{5^k} + \cdots$

Find a. The series starts at $k = 0$:

$$a = \frac{(-1)^0 \cdot 6}{5^0} = \frac{1 \cdot 6}{1} = 6.$$

Find r. The second term ($k = 1$) is $\dfrac{(-1)^1 \cdot 6}{5} = -\dfrac{6}{5}$, so

$$r = \frac{-6/5}{6} = -\frac{1}{5}.$$

Check: $\dfrac{u_{k+1}}{u_k} = \dfrac{(-1)^{k+1}\, 6/5^{k+1}}{(-1)^k\, 6/5^k} = (-1) \cdot \dfrac{5^k}{5^{k+1}} = -\dfrac{1}{5}$. ✓
(The alternating signs give a *negative* ratio.)

Converge or diverge? $|r| = \frac{1}{5} < 1$, so the series converges and

$$\sum_{k=0}^{\infty} \frac{(-1)^k\, 6}{5^k} = \frac{a}{1-r} = \frac{6}{1-\left(-\frac{1}{5}\right)} = \frac{6}{\frac{6}{5}} = 5.$$

Answer: converges, sum = 5

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3. $\displaystyle\sum_{k=1}^{\infty} 2^{2k}\, 3^{1-k} = 4 + \frac{16}{3} + \frac{64}{9} + \cdots + 2^{2k}\, 3^{1-k} + \cdots$

Find a. The series starts at $k = 1$:

$$a = 2^{2(1)} \cdot 3^{1-1} = 2^2 \cdot 3^0 = 4 \cdot 1 = 4.$$

Find r. The second term ($k = 2$) is $2^4 \cdot 3^{-1} = \dfrac{16}{3}$, so

$$r = \frac{16/3}{4} = \frac{16}{12} = \frac{4}{3}.$$

Check (concealed form): $2^{2k}\, 3^{1-k} = \dfrac{4^k}{3^{k-1}} = 4 \cdot \dfrac{4^{k-1}}{3^{k-1}} = 4\left(\dfrac{4}{3}\right)^{k-1}$, which is ar^{k-1} with $a = 4$, $r = \frac{4}{3}$. ✓

Converge or diverge? $|r| = \frac{4}{3} \geq 1$, so the series diverges.

Answer: diverges

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4. $\displaystyle\sum_{k=1}^{\infty} \frac{5^{k+1}}{8^k} = \frac{25}{8} + \frac{125}{64} + \frac{625}{512} + \cdots + \frac{5^{k+1}}{8^k} + \cdots$

Find a. The series starts at $k = 1$:

$$a = \frac{5^{1+1}}{8^1} = \frac{25}{8}.$$

Find r. The second term ($k = 2$) is $\dfrac{5^3}{8^2} = \dfrac{125}{64}$, so

$$r = \frac{125/64}{25/8} = \frac{125}{64} \cdot \frac{8}{25} = \frac{5}{8}.$$

Check: $\dfrac{u_{k+1}}{u_k} = \dfrac{5^{k+2}/8^{k+1}}{5^{k+1}/8^k} = \dfrac{5^{k+2}}{5^{k+1}} \cdot \dfrac{8^k}{8^{k+1}} = 5 \cdot \dfrac{1}{8} = \dfrac{5}{8}$. ✓

Converge or diverge? $|r| = \frac{5}{8} < 1$, so the series converges and

$$\sum_{k=1}^{\infty} \frac{5^{k+1}}{8^k} = \frac{a}{1-r} = \frac{\frac{25}{8}}{1-\frac{5}{8}} = \frac{\frac{25}{8}}{\frac{3}{8}} = \frac{25}{3}.$$

Answer: $\boxed{\text{converges, sum} = \dfrac{25}{3}}$

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5. $\displaystyle\sum_{k=2}^{\infty} \frac{3^k}{2^{3k-1}} = \frac{9}{32} + \frac{27}{256} + \frac{81}{2048} + \cdots + \frac{3^k}{2^{3k-1}} + \cdots$

Find a. Careful: this series starts at $k = 2$, so the first term is

$$a = \frac{3^2}{2^{3(2)-1}} = \frac{9}{2^5} = \frac{9}{32}.$$

(A common mistake is to use $k = 1$ or $k = 0$ here.)

Find r. The second term ($k = 3$) is $\dfrac{3^3}{2^8} = \dfrac{27}{256}$, so

$$r = \frac{27/256}{9/32} = \frac{27}{256} \cdot \frac{32}{9} = \frac{3}{8}.$$

Check: $\dfrac{u_{k+1}}{u_k} = \dfrac{3^{k+1}/2^{3k+2}}{3^k/2^{3k-1}} = \dfrac{3^{k+1}}{3^k} \cdot \dfrac{2^{3k-1}}{2^{3k+2}} = 3 \cdot \dfrac{1}{2^3} = \dfrac{3}{8}$. ✓

Converge or diverge? $|r| = \frac{3}{8} < 1$, so the series converges and

$$\sum_{k=2}^{\infty} \frac{3^k}{2^{3k-1}} = \frac{a}{1-r} = \frac{\frac{9}{32}}{1-\frac{3}{8}} = \frac{\frac{9}{32}}{\frac{5}{8}} = \frac{9}{32} \cdot \frac{8}{5} = \frac{9}{20}.$$

Answer: $\boxed{\text{converges, sum} = \dfrac{9}{20}}$

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6. $3 + \dfrac{3}{4} + \dfrac{3}{4^2} + \cdots + \dfrac{3}{4^{k-1}} + \cdots$

Find a. Read the first term directly from the list: $a = 3$.
(In sigma notation the series is $\sum_{k=1}^{\infty} \frac{3}{4^{k-1}}$, and $k = 1$ gives $\frac{3}{4^0} = 3$.)

Find r. The second term is $\dfrac{3}{4}$, so

$$r = \frac{3/4}{3} = \frac{1}{4}.$$

Check: $\dfrac{3}{4^{k-1}} = 3\left(\dfrac{1}{4}\right)^{k-1}$, which is ar^{k-1} with $a = 3$, $r = \frac{1}{4}$. ✓

Converge or diverge? $|r| = \frac{1}{4} < 1$, so the series converges and

$$\frac{a}{1-r} = \frac{3}{1-\frac{1}{4}} = \frac{3}{\frac{3}{4}} = 4.$$

Answer: $\boxed{\text{converges, sum} = 4}$

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7. $\dfrac{2}{9} - \dfrac{4}{27} + \dfrac{8}{81} - \cdots + \dfrac{(-1)^{k-1}\,2^k}{3^{k+1}} + \cdots$

Find a. Read the first term directly: $a = \dfrac{2}{9}$.
(With the general term starting at $k = 1$: $\frac{(-1)^0\,2^1}{3^2} = \frac{2}{9}$.)

Find r. The second term is $-\dfrac{4}{27}$, so

$$r = \frac{-4/27}{2/9} = -\frac{4}{27} \cdot \frac{9}{2} = -\frac{2}{3}.$$

Check with the third term: $\dfrac{8/81}{-4/27} = \dfrac{8}{81} \cdot \left(-\dfrac{27}{4}\right) = -\dfrac{2}{3}$. ✓

In general, $\dfrac{(-1)^{k-1}\,2^k}{3^{k+1}} = \dfrac{2}{9}\left(-\dfrac{2}{3}\right)^{k-1}$.

Converge or diverge? $|r| = \frac{2}{3} < 1$, so the series converges and

$$\frac{a}{1-r} = \frac{\frac{2}{9}}{1-\left(-\frac{2}{3}\right)} = \frac{\frac{2}{9}}{\frac{5}{3}} = \frac{2}{9} \cdot \frac{3}{5} = \frac{2}{15}.$$

Answer: $\boxed{\text{converges, sum} = \dfrac{2}{15}}$

Type B: Telescoping Sums

Method: (1) Use partial fractions (or log rules) to write each term as a difference $u_k = b_k - b_{k+m}$. (2) Write out s_n and cancel the inner terms to get a closed form. (3) The sum of the series is $\lim_{n\to+\infty} s_n$; if the limit does not exist, the series diverges.

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8. $\displaystyle\sum_{k=1}^{\infty} \frac{1}{k^2+3k+2} = \frac{1}{6} + \frac{1}{12} + \frac{1}{20} + \cdots + \frac{1}{k^2+3k+2} + \cdots$

Partial fractions. $k^2+3k+2 = (k+1)(k+2)$, and

$$\frac{1}{(k+1)(k+2)} = \frac{1}{k+1} - \frac{1}{k+2}.$$

Partial sum.

$$s_n = \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \cdots + \left(\frac{1}{n+1} - \frac{1}{n+2}\right) = \frac{1}{2} - \frac{1}{n+2}.$$

Limit. $\displaystyle\lim_{n\to+\infty} s_n = \lim_{n\to+\infty}\left(\frac{1}{2} - \frac{1}{n+2}\right) = \frac{1}{2}.$

Answer: $\boxed{\text{converges, sum} = \frac{1}{2}}$

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9. $\displaystyle\sum_{k=1}^{\infty} \frac{1}{(2k-1)(2k+1)} = \frac{1}{1\cdot 3} + \frac{1}{3\cdot 5} + \frac{1}{5\cdot 7} + \cdots$

Partial fractions.

$$\frac{1}{(2k-1)(2k+1)} = \frac{1}{2}\left(\frac{1}{2k-1} - \frac{1}{2k+1}\right).$$

Partial sum.

$$s_n = \frac{1}{2}\left[\left(1 - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{5}\right) + \cdots + \left(\frac{1}{2n-1} - \frac{1}{2n+1}\right)\right] = \frac{1}{2}\left(1 - \frac{1}{2n+1}\right).$$

Limit. $\displaystyle\lim_{n\to+\infty} s_n = \frac{1}{2}(1-0) = \frac{1}{2}.$

Answer: $\boxed{\text{converges, sum} = \frac{1}{2}}$

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10. $\displaystyle\sum_{k=1}^{\infty} \ln\left(\frac{k}{k+1}\right) = \ln\frac{1}{2} + \ln\frac{2}{3} + \ln\frac{3}{4} + \cdots$

Log rule. $\ln\left(\dfrac{k}{k+1}\right) = \ln k - \ln(k+1)$.

Partial sum.

$$s_n = (\ln 1 - \ln 2) + (\ln 2 - \ln 3) + \cdots + (\ln n - \ln(n+1)) = \ln 1 - \ln(n+1) = -\ln(n+1).$$

Limit. $\lim\limits_{n\to+\infty} s_n = \lim\limits_{n\to+\infty} (-\ln(n+1)) = -\infty$, so the limit does not exist.

Answer: $\boxed{\text{diverges}}$

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11. $\displaystyle\sum_{k=1}^{\infty} \frac{2}{k(k+2)} = \frac{2}{1\cdot 3} + \frac{2}{2\cdot 4} + \frac{2}{3\cdot 5} + \cdots$

Partial fractions.

$$\frac{2}{k(k+2)} = \frac{1}{k} - \frac{1}{k+2}.$$

Partial sum. Now terms cancel *two apart*, so two positive terms survive at the start and two negative terms at the end:

$$\begin{aligned} s_n &= \left(1 - \frac{1}{3}\right) + \left(\frac{1}{2} - \frac{1}{4}\right) + \left(\frac{1}{3} - \frac{1}{5}\right) + \cdots + \left(\frac{1}{n-1} - \frac{1}{n+1}\right) + \left(\frac{1}{n} - \frac{1}{n+2}\right) \\ &= 1 + \frac{1}{2} - \frac{1}{n+1} - \frac{1}{n+2}. \end{aligned}$$

Limit. $\lim\limits_{n\to+\infty} s_n = 1 + \dfrac{1}{2} - 0 - 0 = \dfrac{3}{2}$.

Answer: $\boxed{\text{converges, sum} = \dfrac{3}{2}}$

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12. $\displaystyle\sum_{k=1}^{\infty} \frac{4}{k^2+4k+3} = \frac{4}{2\cdot 4} + \frac{4}{3\cdot 5} + \frac{4}{4\cdot 6} + \cdots$

Partial fractions. $k^2 + 4k + 3 = (k+1)(k+3)$, and

$$\frac{4}{(k+1)(k+3)} = \frac{2}{k+1} - \frac{2}{k+3}.$$

Partial sum. Terms cancel two apart:

$$\begin{aligned} s_n &= 2\left[\left(\frac{1}{2} - \frac{1}{4}\right) + \left(\frac{1}{3} - \frac{1}{5}\right) + \left(\frac{1}{4} - \frac{1}{6}\right) + \cdots + \left(\frac{1}{n+1} - \frac{1}{n+3}\right)\right] \\ &= 2\left(\frac{1}{2} + \frac{1}{3} - \frac{1}{n+2} - \frac{1}{n+3}\right). \end{aligned}$$

Limit. $\lim\limits_{n\to+\infty} s_n = 2\left(\dfrac{1}{2} + \dfrac{1}{3}\right) = 2\cdot\dfrac{5}{6} = \dfrac{5}{3}$.

Answer: $\boxed{\text{converges, sum} = \frac{5}{3}}$

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13. $\dfrac{1}{4\cdot 5}+\dfrac{1}{5\cdot 6}+\dfrac{1}{6\cdot 7}+\cdots+\dfrac{1}{(k+3)(k+4)}+\cdots$

Sigma form. This is $\displaystyle\sum_{k=1}^{\infty}\frac{1}{(k+3)(k+4)}$ (check: $k=1$ gives $\frac{1}{4\cdot 5}$).

Partial fractions.

$$\frac{1}{(k+3)(k+4)}=\frac{1}{k+3}-\frac{1}{k+4}.$$

Partial sum.

$$s_n=\left(\frac{1}{4}-\frac{1}{5}\right)+\left(\frac{1}{5}-\frac{1}{6}\right)+\cdots+\left(\frac{1}{n+3}-\frac{1}{n+4}\right)=\frac{1}{4}-\frac{1}{n+4}.$$

Limit. $\lim\limits_{n\to+\infty} s_n=\dfrac{1}{4}$.

Answer: $\boxed{\text{converges, sum} = \frac{1}{4}}$

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14. $\dfrac{1}{1\cdot 4}+\dfrac{1}{4\cdot 7}+\dfrac{1}{7\cdot 10}+\cdots+\dfrac{1}{(3k-2)(3k+1)}+\cdots$

Sigma form. This is $\displaystyle\sum_{k=1}^{\infty}\frac{1}{(3k-2)(3k+1)}$ (check: $k=1$ gives $\frac{1}{1\cdot 4}$).

Partial fractions. The factors differ by 3, so

$$\frac{1}{(3k-2)(3k+1)}=\frac{1}{3}\left(\frac{1}{3k-2}-\frac{1}{3k+1}\right).$$

Partial sum.

$$s_n=\frac{1}{3}\left[\left(1-\frac{1}{4}\right)+\left(\frac{1}{4}-\frac{1}{7}\right)+\cdots+\left(\frac{1}{3n-2}-\frac{1}{3n+1}\right)\right]=\frac{1}{3}\left(1-\frac{1}{3n+1}\right).$$

Limit. $\lim\limits_{n\to+\infty} s_n=\dfrac{1}{3}(1-0)=\dfrac{1}{3}$.

Answer: $\boxed{\text{converges, sum} = \frac{1}{3}}$